
NSF FUTURE MANUFACTURING DATA CHALLENGE | FINAL SUBMISSION

Predicting local DED track width from thermal history

Audited physical alignment, local boundaries, uncertainty, and held-out evaluation

Team Submission

July 31, 2026 | github.com/alphons3t/nsf-chuds

Can thermal history predict local geometry after printing?

- **Directed Energy Deposition (DED)** builds metal parts track by track — but final geometry varies along every track.
- In-situ sensors watch the melt pool in real time; **post-process metrology** reveals what was actually built — hours later.
- The challenge: connect what the camera **sees during printing** to the **local geometric variation** measured after printing.
- If we can do this, we unlock **real-time quality prediction and closed-loop control** for metal additive manufacturing.

OUR TASK

Predict local width and boundary position from thermal history

Ground truth: leveled 3D profilometer height maps

Inputs: thermal video + SEM substrate imagery

Scope: 4 tracks spanning different laser powers & scan speeds

Three modalities, one physical x-axis, one held-out condition

THERMAL VIDEO

400 active frames per track

- 400 frames per track · 50 fps
- 14 μm /pixel · 5.6 × 5.6 mm field of view
- Captures melt pool length, width, area
- Lives in the TIME domain

SEM IMAGERY

Substrate-only texture candidate

- Tiled .tif images along each track
- Reveals gouges, texture & roughness
- Used only outside the track (no leakage)
- Lives in the TEXTURE domain

3D PROFILOMETRY

Local width and boundaries

- Bruker/Wyko full-field height maps
- x, y in mm · z resolved to nanometers
- Defines the true deposited track height
- Lives in the SPACE domain

Evaluation is cross-condition: grouped leave-one-track-out development on 8, 10, 14; Track 21 is scored after the pipeline is locked.

From raw frames to leakage-safe local predictions

1**Extract**

Detrend each cross-section and find connected left/right crossings

**2****Align**

Use physical x in mm; never normalized row position

**3****Select**

Compare fixed model families by grouped development CV

**4****Test**

Fit Tracks 8, 10, 14; evaluate width and boundaries on Track 21

Why this matters: every downstream feature is expressed in physical millimeter coordinates along the track — the model learns physics, not sensor artifacts.

Generative AI (LLM) was used transparently as a collaborative coding tool: designing the leveling math, debugging dataframe alignment, and refining visualizations.

Ground truth first: robust boundaries, then physical alignment

STEP 1 - LOCAL GEOMETRY EXTRACTION

Measure one connected bead in every valid cross-section

- Raw profilometer scans carry significant **planar tilt**.
- We isolate the **bare substrate edges** (non-track regions) and fit a 2D plane via linear regression.
- Subtracting the fitted plane levels the substrate to ≈ 0 , exposing the **true deposited height** — our ground truth.

STEP 2 - PHYSICAL TIME-SPACE ALIGNMENT

Match each thermal frame to measured x in millimeters

- Melt pool **length, width & area** extracted from each thermal frame by intensity thresholding (value = 2000).
- Constant scan speed (10 mm/s @ 50 fps) → each frame advances **0.2 mm** along the track.
- Linear interpolation maps all 400 frames onto the **20–100 mm coordinates** of the leveled height profile.

Model selection: rich thermal history wins without SEM

THERMAL DESCRIPTORS

Short histories describe pool shape, energy, and change

- **Rolling averages** of melt pool dimensions over 5- and 10-frame windows.
- These act as a proxy for **heat accumulation and cooling rate** at each point along the path.
- Captures the longitudinal variation a single frame cannot see.

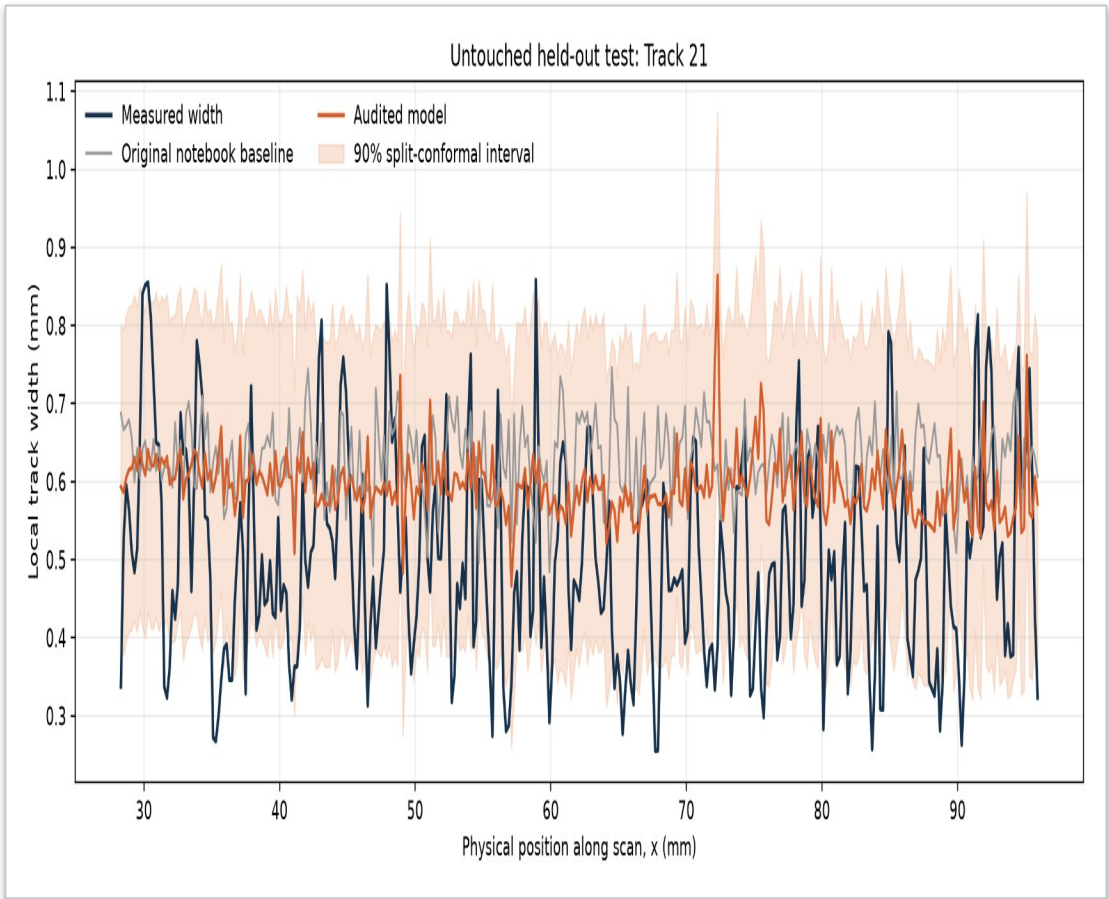
GROUPED MODEL SELECTION

Three development tracks choose the model without Track 21

- Localized **SEM intensity and SEM roughness** matrices extracted from masked SEM imagery.
- Quantifies the **topographical foundation** prior to deposition — gouges, texture, reflectivity.
- Track region masked out to strictly avoid output leakage.

Audited model reduces held-out Track 21 MAE by 12.3%

EXHIBIT 1 - HELD-OUT LOCAL TRACK WIDTH



139 μm

Held-out MAE

12.3% lower

vs. notebook baseline

Baseline MAE: 0.159 mm; audited Ridge MAE: 0.139 mm.

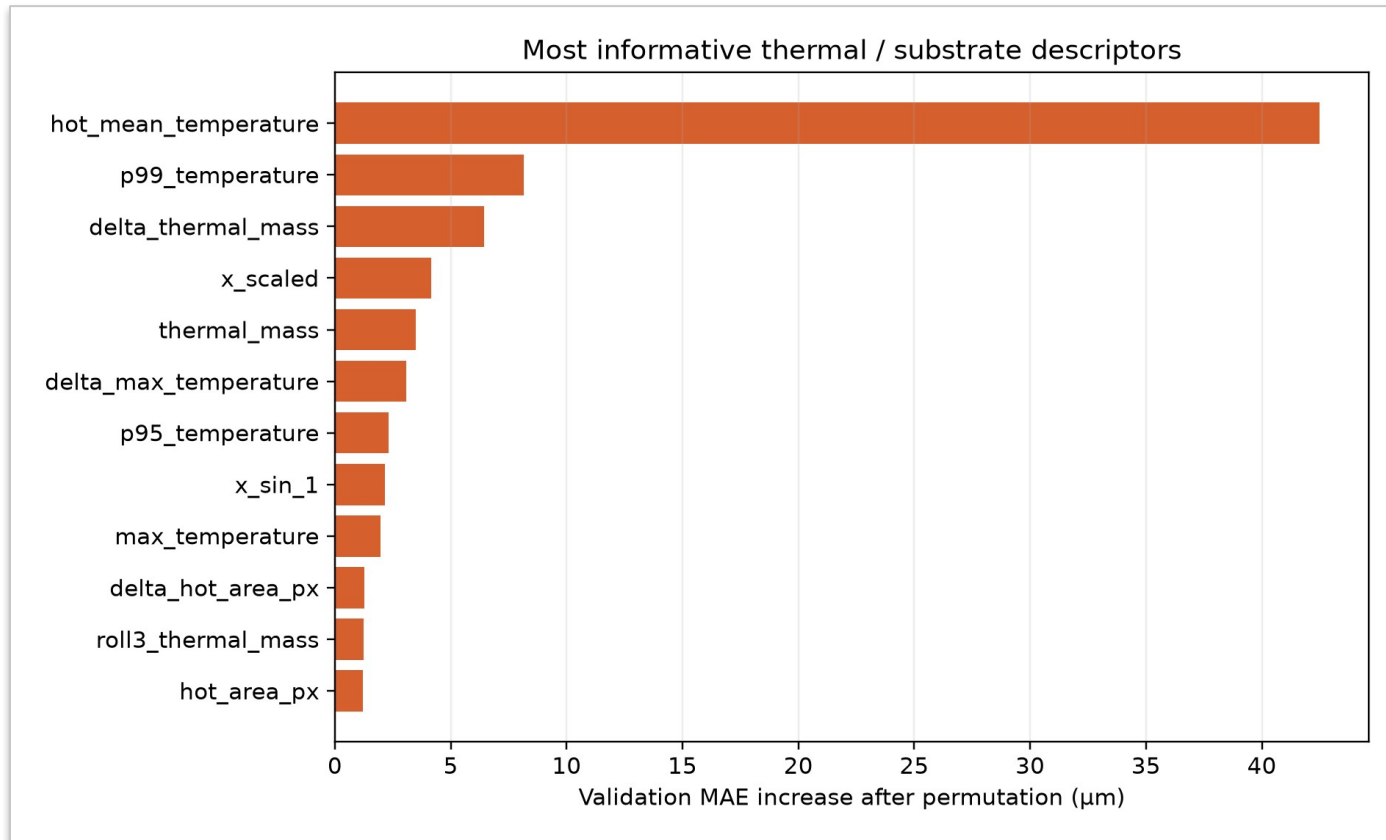
RMSE improves from 0.187 mm to 0.167 mm.

Development grouped-CV MAE improves from 0.137 mm to 0.088 mm.

Point prediction improves, but held-out R^2 remains negative (-0.58).

Thermal history generalizes better than masked SEM

EXHIBIT 2 - VALIDATION PERMUTATION IMPORTANCE



WHAT THE AUDIT FOUND

Hot-region temperature and thermal mass lead

1. Substrate texture governs heat dissipation and liquid metal flow.
2. Thermal history (rolling melt pool areas) ranks next — heat buildup matters.
3. Instantaneous melt pool size alone is the weakest signal.

Improved, but not reliable enough for closed-loop control

Held-out R^2 is -0.58: the condition shift still dominates local fit.

Nominal 90% intervals cover 76.5% of Track 21 points.

Track 21 profilometry is incomplete; only valid regions are scored.

Mean left/right boundary MAE is 0.174 mm.

Four conditions cannot cleanly separate substrate from process.

WHAT WOULD CLOSE THE GAP

Add more powers, plates, repeats, and substrate pre-scans.

Register SEM before processing instead of inferring substrate from post-scan tiles.

Calibrate intervals by laser-power regime or hierarchical conformal methods.

Predict a full contour distribution, not only center plus width.

The audit changed both the model and the claim

1

Better held-out width

Track 21 MAE improves from 159 to 139 μm ;
RMSE improves from 187 to 167 μm .

2

Leakage controls matter

Connected boundaries, physical x alignment,
grouped CV, and a masked-SEM ablation
make the result defensible.

3

Uncertainty remains open

Coverage is 76.5% and R^2 remains negative,
so this is a better benchmark - not a
deployment-ready controller.

“A smaller honest result is more useful than a larger unsupported one.”

Questions? | github.com/alphons3t/nsf-chuds | github.com/joeyperez1-debug/nsfsubmit